Semantic Communication + AI

Can a camera send only the information a receiver needs—without sending every pixel perfectly?

EDGE CAMERA	→	LIMITED + NOISY LINK	→	RECEIVER AI
observes an image		cannot carry everything reliably		only needs the correct class

THESIS

Train the sender and receiver together so the information needed for classification survives a limited, noisy link.

A simulation-first capstone with an optional real-radio demonstration later.

What problem are we solving?

Imagine a remote camera sending images to a server that decides what it sees.

REAL-WORLD TENSION

1. **The sensor is constrained** — Bandwidth, latency and energy are limited at the edge.
2. **The wireless link is imperfect** — Noise can corrupt a short message or make decoding fail.
3. **The receiver has a narrow goal** — For classification, the final question is often simply: what is in the image?

DESIGN QUESTION

Why spend the entire link budget reconstructing every pixel if the receiver ultimately needs a task decision?

The project tests this question; it does not assume the learned method must win.

Goal: preserve the receiver's classification decision without requiring a perfect copy of every image.

How the proposed system changes the usual approach

CONVENTIONAL COMMUNICATION

Compress the image → protect the bits → reconstruct the file → classify it

Primary objective: recover source bits and pixels

PROPOSED LEARNED LINK

Learned sender → noisy link → learned receiver → class decision

Primary objective: preserve information needed for the task

EXPECTED SIGNATURE

As the link worsens, a separated chain may hit a decoding cliff; a learned joint system may degrade more gradually.

IMPORTANT CONTROL

A digital system can also send learned features—so the experiment includes that third arm.

Research question: does the task-oriented joint design help after bandwidth, noise and comparison strength are matched fairly?

Literature survey and research gap

Finite blocklength

[1–4]

Rate backs off from capacity; source and channel dispersions interact.

OPEN ISSUE Need measured practical short-packet loss.

Learned compression

[11–17]

End-to-end transforms and entropy priors improve rate–distortion.

OPEN ISSUE Reconstruction optimization is not task optimization.

DeepJSCC

[5–10,24]

Continuous learned mappings exhibit graceful degradation under mismatch.

OPEN ISSUE Most image studies foreground reconstruction quality.

Task-oriented communication

[18–23]

Learned representations can preserve remote inference under link constraints.

OPEN ISSUE Task awareness must be separated from joint coding.

Gap carried forward: separate neural representation learning from joint channel coding in a controlled three-way comparison.

Objectives and research hypotheses

OBJECTIVES

1. Train a residual neural encoder and dual-head decoder through AWGN.
2. Implement a standards-derived, non-strawman JPEG 2000 + NR LDPC baseline.
3. Tune classical quality, code rate and modulation on validation only.
4. Add a shared-encoder digital AI feature control for attribution.
5. Evaluate paired image outcomes once on the sealed test split.
6. Report positive, null or negative outcomes under one frozen protocol.

H1 · PRIMARY

Low-SNR separation: learned − classical_adaptive

A point qualifies when $\sqrt{N}\cdot\hat{\Delta}(s)/\hat{\sigma}(s) > 1.96$. H1 is supported only if the longest run R_obs contains at least three consecutive qualifying points at or below the training SNR and the calibrated run p-value ≤ 0.05.

Effect size of record: mean paired accuracy difference over the full low-SNR region.

H2	Graceful vs cliff	Fixed classical curve drops ≥30 pp while learned drops ≤15 pp over the frozen window.
H3	Convergence	Paired gap trends toward zero across SNR; crossover is reported, not required.
H4	Attribution	Credit joint coding only if DJSCC also exceeds the task-aware digital control under the H1 rule.

Proposed methodology: three-system comparison

IMAGENETTE-160

canonical
160 × 160
RGB
preprocessing

stable
sample ID
+ split
manifest

CLASSICAL + AI
JPEG 2000
packetise + CRC
NR LDPC + modulation
→ AWGN keyed noise → reconstruct image
frozen ResNet-18

AI FEATURE CONTROL
shared neural features
quantize
same LDPC + modulation
→ AWGN keyed noise → decode features
shared task head

NEURAL DJSCC
residual CNN encoder
power normalize
→ AWGN keyed noise → dual-head decoder
reconstruction + logits

TEST SEALED → G-12

Per-image paired row: verdict + exact k + measured E_s + noise identity + task outcome

How we keep the comparison fair

Equal resource — Same complex-symbol budget k and E_s/N_0 definition; realized symbol energy is logged.

Strong comparator — JPEG 2000 quality, LDPC rate and BPSK/QPSK/16-QAM tune per SNR on validation.

Exact accounting — Container, CRC, code-block and filler bytes count; decode failures remain in the denominator.

Measured BLER — Every required physical identity is characterized; missing evidence is never treated as zero BLER.

Outage policy — Constant-class fallback is measured from validation counts—100/1,000—not assumed as 1/10.

Classifier control — Clean and artifact-finetuned classifier passes separate codec shift from link failure.

Attribution arm — Quantized learned features traverse the same digital chain, isolating task awareness from joint coding.

One test campaign — All choices freeze before G-12; paired inference retains complete image × system trajectories.

Fail-closed rule: uncharacterized, infeasible and failed cases are explicit outcomes—not silently dropped rows.

Preliminary work and feasibility

NO HEADLINE COMPARISON EXISTS YET · bounded smoke is integration evidence, not scientific outcome evidence

G-1 · ResNet-18

89.8%

trained from scratch
898/1,000 validation top-1 · PASS

G-7 · neural model

1.64 M

residual DJSCC parameters · batch 32
48.68 s/epoch · 1.004 GiB VRAM

G-2 · digital PHY

≤ 0.0037 dB

observed waterfall displacement
all three modulations within 0.5 dB · PASS

W4 · integration

end-to-end

JPEG 2000 → LDPC → AWGN → decode
records + verifier · bounded PASS

G8_C · CURRENT

C2 paused

authenticated durable checkpoint
source epoch 2 registered · safely resumable

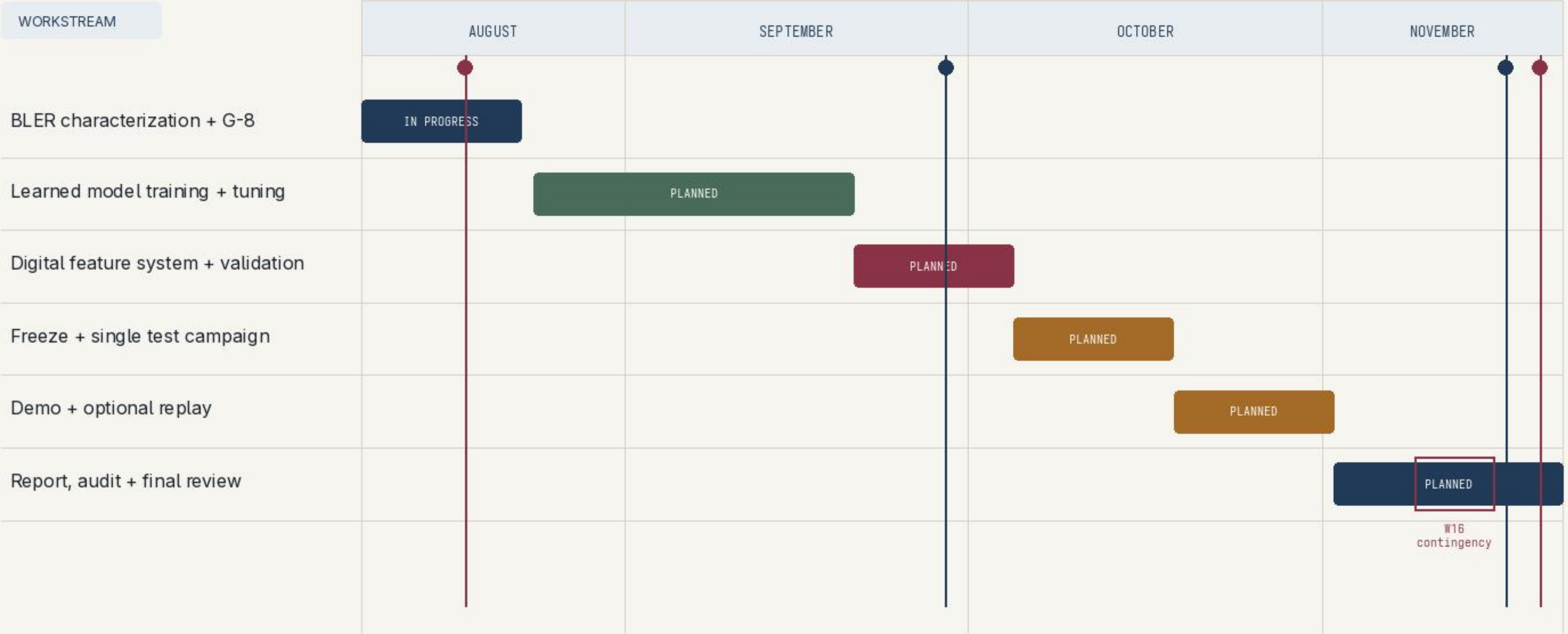
BLER table: not frozen
selection / inference / training / test: 0

Interpretation: the reference task, computational budget and digital chain are verified; the comparison itself remains ahead.

SUBJECT KNOWLEDGE

Authenticated records: G-1, G-2, G-7, W4 and G8_C · test split access = 0

Project Gantt chart



Review dates: 18–22 Aug · 29 Sep–3 Oct · 17–21 Nov · report due 20 Nov

Only BLER characterization is in progress; all later work is planned.

Edge AI application, AIML contribution and scope

APPLICATION

Edge AI application

Remote cameras and sensors may need receiver-side inference over limited, noisy links.

The sender runs a compact encoder; the receiver owns the decoder and task head. This represents split edge/cloud inference.

The measured task is Imagenette-160 classification, but the method is neural representation learning for communication.

IMPLEMENTATION

Deployment ladder

TIER 1 · required
Simulation-first, offline and reproducible.

TIER 2 · stretch after G-5
Conducted SDR I/Q replay; prerecorded outcome expected.

TIER 3 · stretch
Edge placement only if latency and procurement gates pass.

Candidate: HackRF One + attenuation chain + RTL-SDR; no purchase yet.

METHOD

AI/ML contribution

Residual CNN encoder with GroupNorm and PReLU.

Dual-head decoder for class logits and reconstruction.

Supervised multi-task loss: cross-entropy + λMSE.

Differentiable AWGN enables end-to-end training.

The digital feature arm isolates representation learning from joint coding.

TECHNICAL BOUNDARY

OpenJPEG 2.5.4 + TS 38.212-derived LDPC/rate matching over abstract AWGN—not a full 5G NR link claim

No guaranteed win, measured energy saving or real-radio generalization; completion means executing the preregistered protocol.

Explicit First Review rubric-to-slide map

CRITERION	SLIDES	EVIDENCE EXPOSED IN THE DECK
Motivation	2–3	Short packets, fixed k, task accuracy, finite-blocklength costs
Objectives	3, 5–7, 10	Neural method, three-arm build, fair protocol and attribution control
Hypothesis	5	Exact H1 run rule; H2–H4 preregistered; crossing not required
Problem Survey	2, 4, 12	30-source synthesis across four literature families
Subject Knowledge	3–8, 10	Residual CNN, multi-task loss, AWGN, controls, evidence and scope
Time Plan	9	Gate-ordered critical path and fixed review/report dates

Final PASS still requires the real PPT review, four-member rehearsal, guide acknowledgement and annotated review-1-basis tag.

Selected references and evidence provenance

SELECTED LITERATURE

[1] Shannon, A Mathematical Theory of Communication, 1948.

[2] Gastpar, Rimoldi & Vetterli, To Code, or Not to Code, 2003.

[3] Polyanskiy, Poor & Verdú, Finite Blocklength, 2010.

[4] Kostina & Verdú, Lossy JSCC at Finite Blocklength, 2013.

[5] Bourtsoulatze, Kurka & Gündüz, DeepJSCC, 2019.

[6] Kurka & Gündüz, DeepJSCC-f, 2020.

[7] Kurka & Gündüz, Bandwidth-Agile DeepJSCC, 2021.

[11] Ballé, Laparra & Simoncelli, Learned Compression, 2017.

[12] Ballé et al., Scale Hyperprior, 2018.

[17] Blau & Michaeli, Perception–Distortion, 2018.

[18] Jankowski, Gündüz & Mikolajczyk, Wireless Retrieval, 2021.

[19] Shao, Mao & Zhang, Task-Oriented Edge Inference, 2021.

[20] Xie et al., Deep Learning Enabled Semantic Communication, 2021.

REPOSITORY EVIDENCE

G-1	`results/reference_classifier/g1_adjudication.json`	89.8% validation; test sealed
G-7	`results/profiling/g7_djscc_profile.json`	1.64M params; measured epoch/VRAM
G-2	`results/baseline/g2/g2_adjudication.json`	golden vectors + BLER conformance
W4	`results/baseline/w4/integration_adjudication.json`	bounded end-to-end integration
G8_C	`results/baseline/g8/campaign_state.json`	authenticated characterization cursor
Spec	`spec/SPEC.md`	normative hypotheses and requirements
Plan	`docs/gantt-plan.md`	dates, gates and current status
Deploy	`docs/deployment-dossier.md`	simulation-first + SDR stretch

Refresh G8_C status from instructions/RESUME.md on submission day.